

▼ Problem Statement



The purpose of this project is to analyze sales data to understand the factors that affect sales and profit. The dataset contains information about products, units sold, sales, profit, discounts, customers, segments, countries, and time periods. The analysis will identify high-performing products, sales and profit trends, and relationships between important variables. The findings will help identify areas for improving sales performance, profitability, and business decision-making.

Import Libraries

```
[6]: import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import numpy as np
```

Matplotlib is building the font cache; this may take a moment.

▼ Load the Dataset ¶

```
[39]: df = pd.read_csv('Dataset.csv')
      print(df)
```

	Segment	Country	Product	Discount	Band	\
0	Government	Germany	Carretera		NaN	
1	Government	Germany	Paseo		NaN	
2	Government	Canada	Paseo		NaN	
3	Government	Germany	Paseo		NaN	
4	Government	Germany	Velo		NaN	
..	...	...	...		...	
695	Midmarket	Canada	Paseo		High	
696	Midmarket	Canada	Paseo		High	
697	Enterprise	Germany	Paseo		High	
698	Midmarket	Germany	Paseo		High	
699	Channel Partners	United States of America	Paseo		High	

	Units Sold	Manufacturing Price	Sale Price	Gross Sales	Discounts	\
0	1513.0	3	350	529550.0	0.00	
1	1006.0	10	350	352100.0	0.00	
2	1725.0	10	350	603750.0	0.00	
3	1513.0	10	350	529550.0	0.00	
4	1006.0	120	350	352100.0	0.00	
..	...	...	...	...	...	
695	1614.0	10	15	24210.0	3631.50	
696	2559.0	10	15	38385.0	5757.75	
697	1085.0	10	125	135625.0	20343.75	
698	1175.0	10	15	17625.0	2643.75	
699	914.0	10	12	10968.0	1645.20	

	Sales	COGS	Profit	Date	Month	Name	Year
0	529550	393380.0	136170.00	01-Dec-14	December		2014
1	352100	261560.0	90540.00	01-Jun-14	June		2014
2	603750	448500.0	155250.00	01-Nov-13	November		2013
3	529550	393380.0	136170.00	01-Dec-14	December		2014
4	352100	261560.0	90540.00	01-Jun-14	June		2014
..	...	...	...	...	...		...
695	20579	16140.0	4438.50	01-Apr-14	April		2014
696	32627	25590.0	7037.25	01-Aug-14	August		2014
697	115281	130200.0	-14918.75	01-Oct-14	October		2014
698	14981	11750.0	3231.25	01-Oct-14	October		2014
699	9323	2742.0	6580.80	01-Dec-14	December		2014

[700 rows x 15 columns]

```
[40]: df.head()
```

```
[40]:
```

	Segment	Country	Product	Discount Band	Units Sold	Manufacturing Price	Sale Price	Gross Sales	Discounts	Sales	COGS	Profit	Date	Month Name	Year
0	Government	Germany	Carretera	NaN	1513.0	3	350	529550.0	0.0	529550	393380.0	136170.0	01-Dec-14	December	2014
1	Government	Germany	Paseo	NaN	1006.0	10	350	352100.0	0.0	352100	261560.0	90540.0	01-Jun-14	June	2014
2	Government	Canada	Paseo	NaN	1725.0	10	350	603750.0	0.0	603750	448500.0	155250.0	01-Nov-13	November	2013
3	Government	Germany	Paseo	NaN	1513.0	10	350	529550.0	0.0	529550	393380.0	136170.0	01-Dec-14	December	2014
4	Government	Germany	Velo	NaN	1006.0	120	350	352100.0	0.0	352100	261560.0	90540.0	01-Jun-14	June	2014

```
[41]: df.head(10)
```

```
[41]:
```

	Segment	Country	Product	Discount Band	Units Sold	Manufacturing Price	Sale Price	Gross Sales	Discounts	Sales	COGS	Profit	Date	Month Name	Year
0	Government	Germany	Carretera	NaN	1513.0	3	350	529550.0	0.0	529550	393380.0	136170.0	01-Dec-14	December	2014
1	Government	Germany	Paseo	NaN	1006.0	10	350	352100.0	0.0	352100	261560.0	90540.0	01-Jun-14	June	2014
2	Government	Canada	Paseo	NaN	1725.0	10	350	603750.0	0.0	603750	448500.0	155250.0	01-Nov-13	November	2013
3	Government	Germany	Paseo	NaN	1513.0	10	350	529550.0	0.0	529550	393380.0	136170.0	01-Dec-14	December	2014
4	Government	Germany	Velo	NaN	1006.0	120	350	352100.0	0.0	352100	261560.0	90540.0	01-Jun-14	June	2014
5	Government	France	VTT	NaN	1527.0	250	350	534450.0	0.0	534450	397020.0	137430.0	01-Sep-13	September	2013
6	Government	France	Amarilla	NaN	2750.0	260	350	962500.0	0.0	962500	715000.0	247500.0	01-Feb-14	February	2014
7	Government	Mexico	Carretera	Low	1210.0	3	350	423500.0	4235.0	419265	314600.0	104665.0	01-Mar-14	March	2014
8	Government	Mexico	Carretera	Low	1397.0	3	350	488950.0	4889.5	484061	363220.0	120840.5	01-Oct-14	October	2014
9	Government	France	Carretera	Low	2155.0	3	350	754250.0	7542.5	746708	560300.0	186407.5	01-Dec-14	December	2014

```
[43]: df.tail()
```

	Segment	Country	Product	Discount Band	Units Sold	Manufacturing Price	Sale Price	Gross Sales	Discounts	Sales	COGS	Profit	Date	Month Name	Year
695	Midmarket	Canada	Paseo	High	1614.0	10	15	24210.0	3631.50	20579	16140.0	4438.50	01-Apr-14	April	2014
696	Midmarket	Canada	Paseo	High	2559.0	10	15	38385.0	5757.75	32627	25590.0	7037.25	01-Aug-14	August	2014
697	Enterprise	Germany	Paseo	High	1085.0	10	125	135625.0	20343.75	115281	130200.0	-14918.75	01-Oct-14	October	2014
698	Midmarket	Germany	Paseo	High	1175.0	10	15	17625.0	2643.75	14981	11750.0	3231.25	01-Oct-14	October	2014
699	Channel Partners	United States of America	Paseo	High	914.0	10	12	10968.0	1645.20	9323	2742.0	6580.80	01-Dec-14	December	2014

```
[44]: df.tail(10)
```

	Segment	Country	Product	Discount Band	Units Sold	Manufacturing Price	Sale Price	Gross Sales	Discounts	Sales	COGS	Profit	Date	Month Name	Year
690	Channel Partners	France	Paseo	High	1393.0	10	12	16716.0	2340.24	14376	4179.0	10196.76	01-Oct-14	October	2014
691	Midmarket	Canada	Paseo	High	2470.0	10	15	37050.0	5187.00	31863	24700.0	7163.00	01-Sep-13	September	2013
692	Midmarket	Canada	Paseo	High	1743.0	10	15	26145.0	3660.30	22485	17430.0	5054.70	01-Oct-13	October	2013
693	Channel Partners	United States of America	Paseo	High	2914.0	10	12	34968.0	4895.52	30072	8742.0	21330.48	01-Oct-14	October	2014
694	Channel Partners	Canada	Paseo	High	2222.0	10	12	26664.0	3732.96	22931	6666.0	16265.04	01-Nov-13	November	2013
695	Midmarket	Canada	Paseo	High	1614.0	10	15	24210.0	3631.50	20579	16140.0	4438.50	01-Apr-14	April	2014
696	Midmarket	Canada	Paseo	High	2559.0	10	15	38385.0	5757.75	32627	25590.0	7037.25	01-Aug-14	August	2014
697	Enterprise	Germany	Paseo	High	1085.0	10	125	135625.0	20343.75	115281	130200.0	-14918.75	01-Oct-14	October	2014
698	Midmarket	Germany	Paseo	High	1175.0	10	15	17625.0	2643.75	14981	11750.0	3231.25	01-Oct-14	October	2014
699	Channel Partners	United States of America	Paseo	High	914.0	10	12	10968.0	1645.20	9323	2742.0	6580.80	01-Dec-14	December	2014

```
[45]: df.iloc[:10]
```

```
[45]:
```

	Segment	Country	Product	Discount Band	Units Sold	Manufacturing Price	Sale Price	Gross Sales	Discounts	Sales	COGS	Profit	Date	Month Name	Year
0	Government	Germany	Carretera	NaN	1513.0	3	350	529550.0	0.0	529550	393380.0	136170.0	01-Dec-14	December	2014
1	Government	Germany	Paseo	NaN	1006.0	10	350	352100.0	0.0	352100	261560.0	90540.0	01-Jun-14	June	2014
2	Government	Canada	Paseo	NaN	1725.0	10	350	603750.0	0.0	603750	448500.0	155250.0	01-Nov-13	November	2013
3	Government	Germany	Paseo	NaN	1513.0	10	350	529550.0	0.0	529550	393380.0	136170.0	01-Dec-14	December	2014
4	Government	Germany	Velo	NaN	1006.0	120	350	352100.0	0.0	352100	261560.0	90540.0	01-Jun-14	June	2014
5	Government	France	VTT	NaN	1527.0	250	350	534450.0	0.0	534450	397020.0	137430.0	01-Sep-13	September	2013
6	Government	France	Amarilla	NaN	2750.0	260	350	962500.0	0.0	962500	715000.0	247500.0	01-Feb-14	February	2014
7	Government	Mexico	Carretera	Low	1210.0	3	350	423500.0	4235.0	419265	314600.0	104665.0	01-Mar-14	March	2014
8	Government	Mexico	Carretera	Low	1397.0	3	350	488950.0	4889.5	484061	363220.0	120840.5	01-Oct-14	October	2014
9	Government	France	Carretera	Low	2155.0	3	350	754250.0	7542.5	746708	560300.0	186407.5	01-Dec-14	December	2014

```
[46]: df.columns
```

```
[46]: Index(['Segment', 'Country', 'Product', 'Discount Band', 'Units Sold',  
        'Manufacturing Price', 'Sale Price', 'Gross Sales', 'Discounts',  
        'Sales', 'COGS', 'Profit', 'Date', 'Month Name', 'Year'],  
       dtype='object')
```

```
[47]: df.columns.to_list()
```

```
[47]: ['Segment',  
      'Country',  
      'Product',  
      'Discount Band',  
      'Units Sold',  
      'Manufacturing Price',  
      'Sale Price',  
      'Gross Sales',  
      'Discounts',  
      'Sales',  
      'COGS',  
      'Profit',  
      'Date',  
      'Month Name',  
      'Year']
```

```
[48]: df.shape
```

```
[48]: (700, 15)
```

```
[49]: print("Number of rows",df.shape[0])  
      print("Number of columns",df.shape[1])
```

```
Number of rows 700  
Number of columns 15
```

```
[50]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>  
RangeIndex: 700 entries, 0 to 699  
Data columns (total 15 columns):  
#   Column                Non-Null Count  Dtype  
---  ---  
0   Segment                700 non-null   object  
1   Country                700 non-null   object  
2   Product                700 non-null   object  
3   Discount Band          647 non-null   object  
4   Units Sold             700 non-null   float64  
5   Manufacturing Price     700 non-null   int64  
6   Sale Price             700 non-null   int64  
7   Gross Sales            700 non-null   float64  
8   Discounts              700 non-null   float64  
9   Sales                  700 non-null   int64  
10  COGS                   700 non-null   float64  
11  Profit                 700 non-null   float64  
12  Date                   700 non-null   object  
13  Month Name             700 non-null   object  
14  Year                   700 non-null   int64  
dtypes: float64(5), int64(4), object(6)  
memory usage: 82.2+ KB
```

[51]: df.dtypes

```
[51]: Segment      object
      Country      object
      Product      object
      Discount Band object
      Units Sold   float64
      Manufacturing Price int64
      Sale Price   int64
      Gross Sales  float64
      Discounts    float64
      Sales        int64
      COGS         float64
      Profit       float64
      Date         object
      Month Name   object
      Year         int64
      dtype: object
```

[52]: df.describe()

```
[52]:
```

	Units Sold	Manufacturing Price	Sale Price	Gross Sales	Discounts	Sales	COGS	Profit	Year
count	700.000000	700.000000	700.000000	7.000000e+02	700.000000	7.000000e+02	700.000000	700.000000	700.000000
mean	1608.294286	96.477143	118.428571	1.827594e+05	13150.354629	1.696091e+05	145475.211429	24133.860371	2013.750000
std	867.427859	108.602612	136.775515	2.542623e+05	22962.928775	2.367264e+05	203865.506118	42760.626563	0.433322
min	200.000000	3.000000	7.000000	1.799000e+03	0.000000	1.655000e+03	918.000000	-40617.500000	2013.000000
25%	905.000000	5.000000	12.000000	1.739175e+04	800.320000	1.592800e+04	7490.000000	2805.960000	2013.750000
50%	1542.500000	10.000000	20.000000	3.798000e+04	2585.250000	3.554050e+04	22506.250000	9242.200000	2014.000000
75%	2229.125000	250.000000	300.000000	2.790250e+05	15956.343750	2.610775e+05	245607.500000	22662.000000	2014.000000
max	4492.500000	260.000000	350.000000	1.207500e+06	149677.500000	1.159200e+06	950625.000000	262200.000000	2014.000000

[53]: df.describe().T

```
[53]:
```

	count	mean	std	min	25%	50%	75%	max
Units Sold	700.0	1608.294286	867.427859	200.0	905.00	1542.50	2229.12500	4492.5
Manufacturing Price	700.0	96.477143	108.602612	3.0	5.00	10.00	250.00000	260.0
Sale Price	700.0	118.428571	136.775515	7.0	12.00	20.00	300.00000	350.0
Gross Sales	700.0	182759.426429	254262.284378	1799.0	17391.75	37980.00	279025.00000	1207500.0
Discounts	700.0	13150.354629	22962.928775	0.0	800.32	2585.25	15956.34375	149677.5
Sales	700.0	169609.121429	236726.363843	1655.0	15928.00	35540.50	261077.50000	1159200.0
COGS	700.0	145475.211429	203865.506118	918.0	7490.00	22506.25	245607.50000	950625.0
Profit	700.0	24133.860371	42760.626563	-40617.5	2805.96	9242.20	22662.00000	262200.0
Year	700.0	2013.750000	0.433322	2013.0	2013.75	2014.00	2014.00000	2014.0

```
[54]: df.isnull().sum()
```

```
[54]: Segment          0
      Country         0
      Product         0
      Discount Band    53
      Units Sold       0
      Manufacturing Price 0
      Sale Price       0
      Gross Sales      0
      Discounts        0
      Sales            0
      COGS             0
      Profit           0
      Date             0
      Month Name       0
      Year             0
      dtype: int64
```

```
[25]: df.duplicated().sum()
```

```
[25]: np.int64(0)
```

Handling Missing Values

```
[55]: df_filled = df.copy()
```

```
df_filled["Units Sold"] = df_filled["Units Sold"].fillna(df_filled["Units Sold"].mean())
print(df_filled["Units Sold"])
```

```
0    1513.0
1    1006.0
2    1725.0
3    1513.0
4    1006.0
```

...

```
695    1614.0
696    2559.0
697    1085.0
698    1175.0
699     914.0
```

Name: Units Sold, Length: 700, dtype: float64

```
[56]: df_filled = df.copy()
```

```
df_filled["Sale Price"] = df_filled["Sale Price"].fillna(df_filled["Sale Price"].mean())
print(df_filled["Sale Price"])
```

```
0     350
1     350
2     350
3     350
4     350
```

...

```
695     15
696     15
697    125
698     15
699     12
```

Name: Sale Price, Length: 700, dtype: int64

```
[57]: df_filled = df.copy()

df_filled["Manufacturing Price"] = df_filled["Manufacturing Price"].fillna(df_filled["Manufacturing Price"].median())
print(df_filled["Manufacturing Price"])
```

```
0      3
1     10
2     10
3     10
4    120
```

...

```
695    10
696    10
697    10
698    10
699    10
```

```
Name: Manufacturing Price, Length: 700, dtype: int64
```

EDA

Which segment has the highest sales?

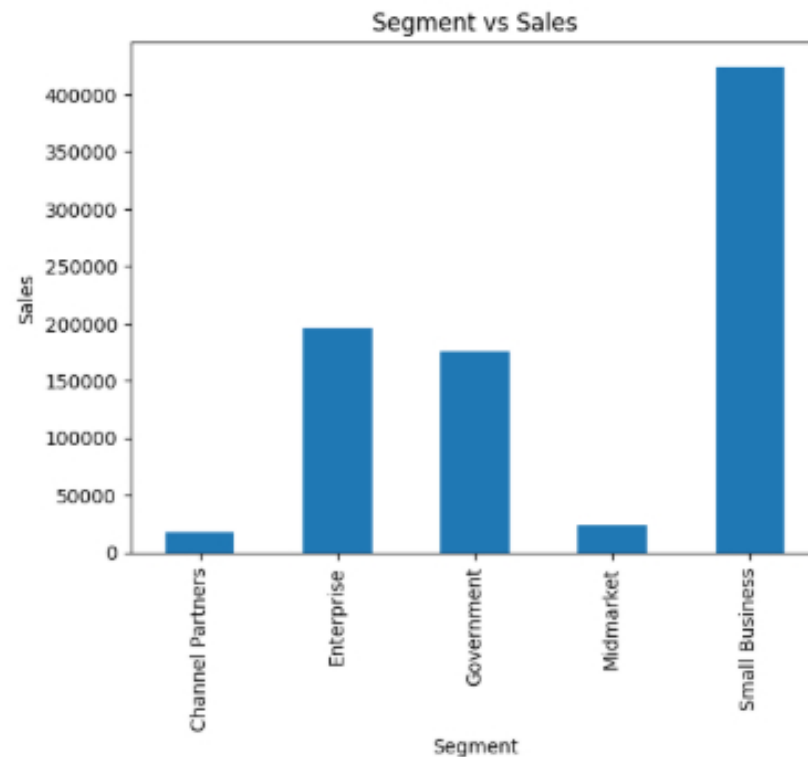
```
[116]: segment = df.groupby("Segment")["Sales"].mean()  
print("Highest Sales of Segment" , segment)
```

```
Highest Sales of Segment Segment  
Channel Partners    18005.940000  
Enterprise          196117.050000  
Government          175014.276667  
Midmarket           23818.830000  
Small Business      424279.200000  
Name: Sales, dtype: float64
```

```
[117]: segment = df.groupby("Segment")["Sales"].mean().idxmax()  
print("Highest Sales Segment" , segment)
```

```
Highest Sales of Segment Small Business
```

```
[132]: segment = df.groupby("Segment")["Sales"].mean().plot.bar()  
plt.title("Segment vs Sales")  
plt.xlabel("Segment")  
plt.ylabel("Sales")  
plt.show()
```



Which country has the lowest sales?

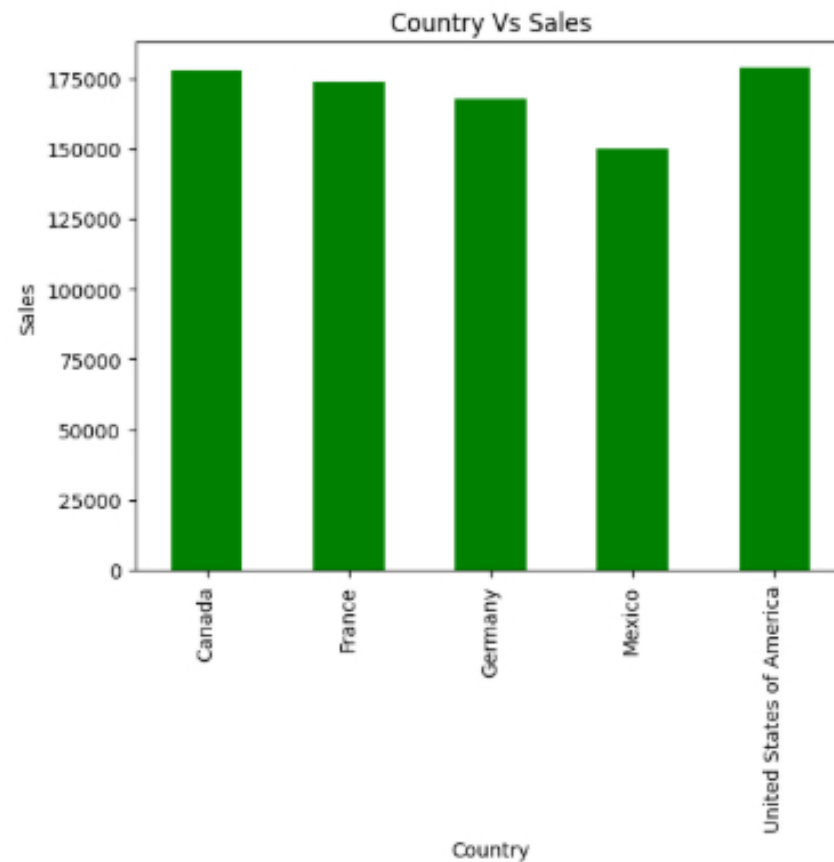
```
[114]: country = df.groupby("Country")["Sales"].mean()  
print("Lowest Sales of Country", country)
```

```
Lowest Sales of Country Country  
Canada                177769.000000  
France                173958.442857  
Germany               167895.371429  
Mexico                149638.264286  
United States of America 178784.528571  
Name: Sales, dtype: float64
```

```
[115]: country = df.groupby("Country")["Sales"].mean().idxmin()  
print("Lowest Sales of Country", country)
```

```
Lowest Sales of Country Mexico
```

```
[133]: country = df.groupby("Country")["Sales"].mean().plot.bar(color="green")  
plt.title("Country Vs Sales")  
plt.xlabel("Country")  
plt.ylabel("Sales")  
plt.show()
```



Which Product has the Highest Sales

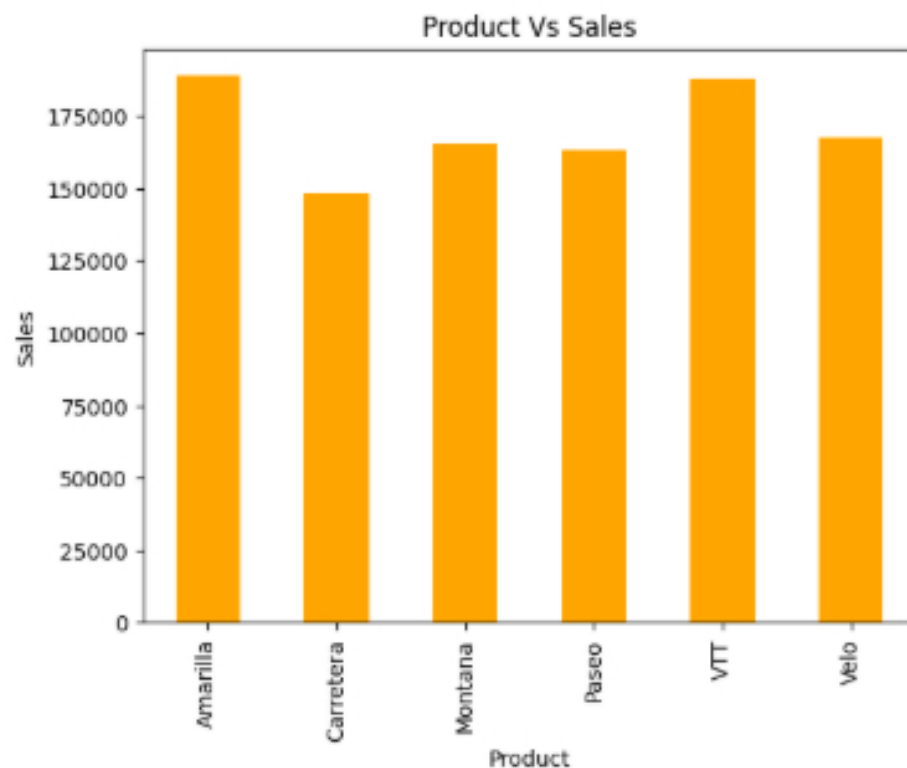
```
[118]: product = df.groupby("Product")["Sales"].mean()  
print("Product with highest sales",product)
```

```
Product with highest sales Product  
Amarilla      188799.127660  
Carretera     148551.752688  
Montana       165492.548387  
Paseo         163421.549505  
VTT           188182.825688  
Velo          167431.798165  
Name: Sales, dtype: float64
```

```
[119]: product = df.groupby("Product")["Sales"].mean().idxmax()  
print("Product with highest sales",product)
```

```
Product with highest sales Amarilla
```

```
[135]: product = df.groupby("Product")["Sales"].mean().plot.bar(color="Orange")  
plt.title("Product Vs Sales")  
plt.xlabel("Product")  
plt.ylabel("Sales")  
plt.show()
```



Which product has highest in the profit

```
[77]: profit = df.groupby("Product")["Profit"].mean()  
print("Product with Highest Profit", profit)
```

Highest Number of Profit in the Product Product
Amarilla 29937.277234
Carretera 19643.063280
Montana 22739.299785
Paseo 23749.692822
VTT 27840.440550
Velo 21155.894174
Name: Profit, dtype: float64

```
[121]: profit = df.groupby("Product")["Profit"].mean().idxmax()  
print("Product with Highest Profit", profit)
```

Product with Highest Profit Amarilla

```
[136]: profit = df.groupby("Product")["Profit"].mean().plot.bar(color="purple")  
plt.title("Product Vs Profit")  
plt.xlabel("Product")  
plt.ylabel("Profit")  
plt.show()
```



Which Year has highest in the Sales

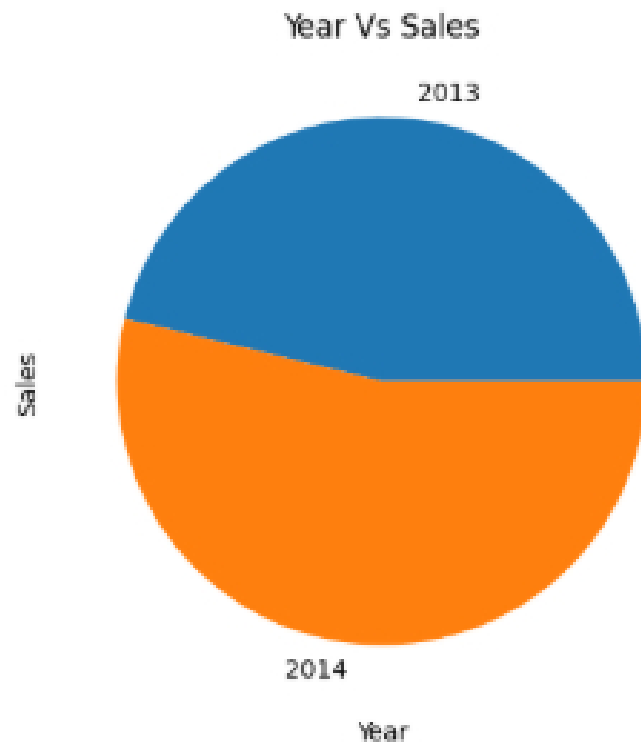
```
[123]: year = df.groupby("Year")["Sales"].mean()  
print("Year with highest sales", year)
```

```
Year with highest sales Year  
2013    158944.388571  
2014    175838.699848  
Name: Sales, dtype: float64
```

```
[124]: year = df.groupby("Year")["Sales"].mean().idxmax()  
print("Year with highest sales", year)
```

```
Year with highest sales 2014
```

```
[137]: year = df.groupby("Year")["Sales"].mean().plot.pie()  
plt.title("Year Vs Sales")  
plt.xlabel("Year")  
plt.ylabel("Sales")  
plt.show()
```



Which product has the highest units sold?

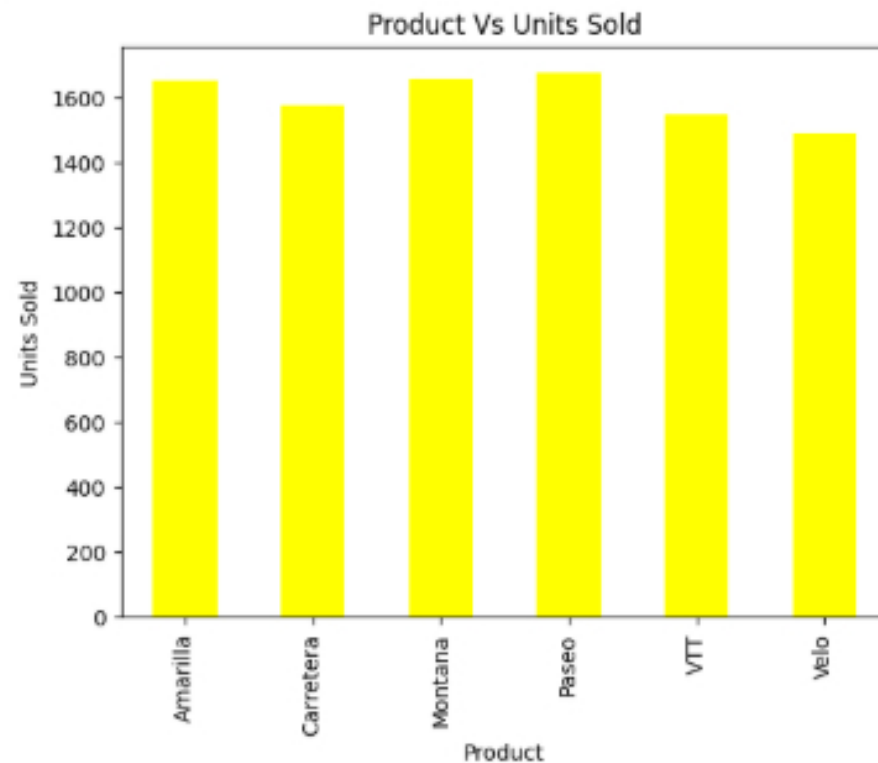
```
[125]: Units_sold = df.groupby("Product")["Units Sold"].mean()  
print("Highest number of units sold:", Units_sold)
```

```
Highest number of units sold: Product  
Amarilla    1652.287234  
Carretera   1578.989247  
Montana     1658.843811  
Paseo       1674.452978  
VTT         1548.467898  
Velo        1498.133828  
Name: Units Sold, dtype: float64
```

```
[126]: Units_sold = df.groupby("Product")["Units Sold"].mean().idxmax()  
print("Highest number of units sold:", Units_sold)
```

```
Highest number of units sold: Paseo
```

```
[139]: Units_sold = df.groupby("Product")["Units Sold"].mean().plot.bar(color="yellow")  
plt.title("Product Vs Units Sold")  
plt.xlabel("Product")  
plt.ylabel("Units Sold")  
plt.show()
```



What are the major sales and profit trends in the dataset?

```
[92]: country = df.groupby("Country")[["Sales", "Profit"]].sum()
print(country)
```

Country	Sales	Profit
Canada	24887660	3529228.885
France	24354182	3781020.780
Germany	23505352	3680388.820
Mexico	20949357	2987523.110
United States of America	25029834	2995540.665

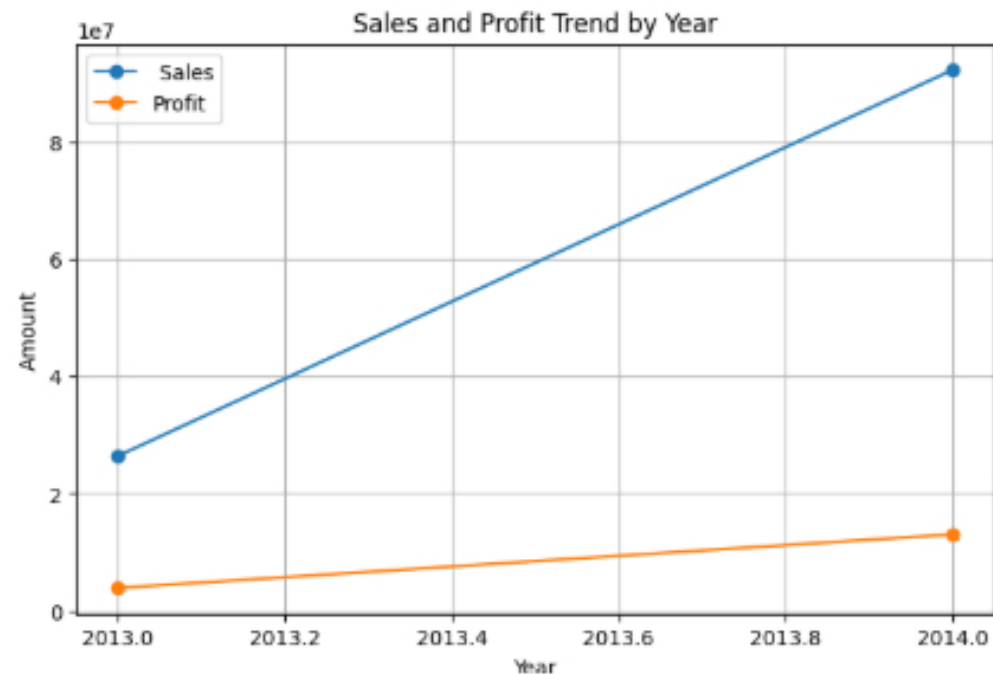
```
[96]: year = df.groupby("Year")[["Sales", "Profit"]].sum()
print(year)
```

	Sales	Profit
Year		
2013	26415268	3878464.51
2014	92311117	13015237.75

```
[108]: import matplotlib.pyplot as plt

year.plot(kind="line", marker="o", figsize=(8, 5))

plt.title("Sales and Profit Trend by Year")
plt.xlabel("Year")
plt.ylabel("Amount")
plt.grid(True)
plt.show()
```



```
[97]: segment = df.groupby("Segment")[["Sales", "Profit"]].sum()
print(segment)
```

	Sales	Profit
Segment		
Channel Partners	1888594	1.316883e+06
Enterprise	19611785	-6.145456e+05
Government	52584283	1.138817e+07
Midmarket	2381883	6.681831e+05
Small Business	42427928	4.143168e+06

```
[98]: product = df.groupby("Product")[["Sales", "Profit"]].sum()
print(product)
```

	Sales	Profit
Product		
Amarilla	17747118	2814184.868
Carretera	13815313	1826884.885
Montana	15398887	2114754.888
Paseo	33811153	4797437.958
VTT	28511928	3834688.828
Velo	18258866	2385992.465

Are there any outliers in Sales, Profit, or Units Sold?

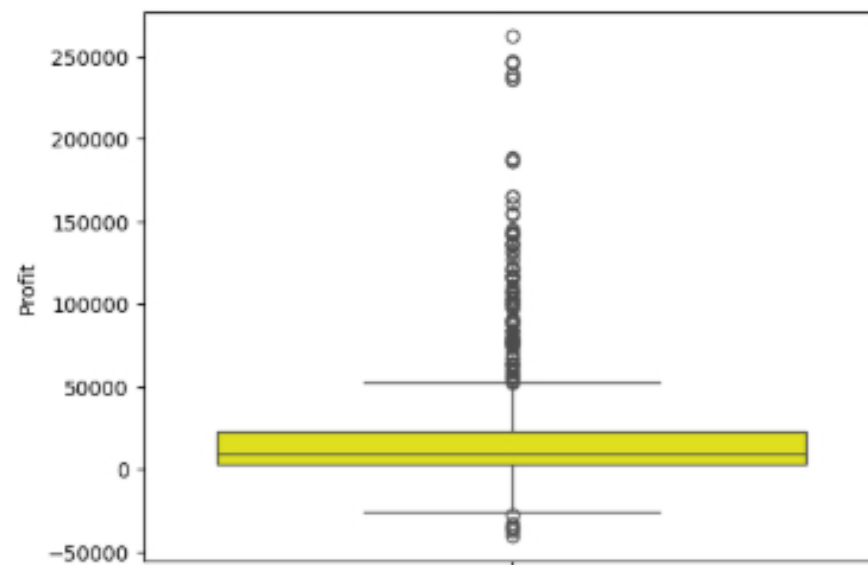
```
[145]: sns.boxplot(y="Sales",data=df,color="Orange")
```

```
[145]: <Axes: ylabel='Sales'>
```



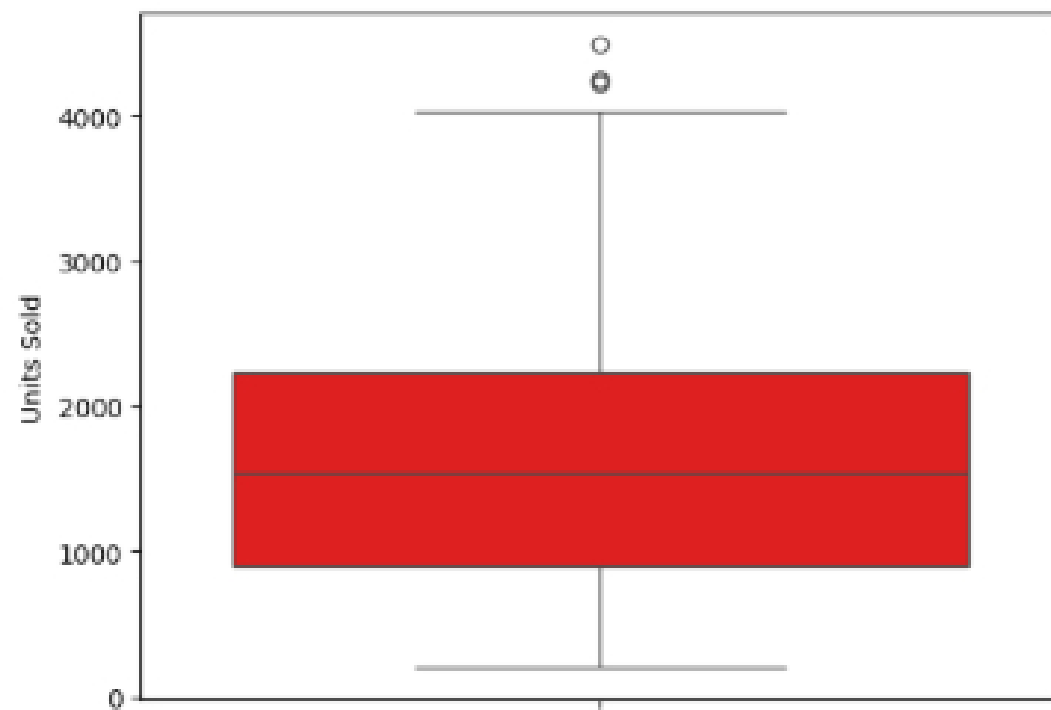
```
[146]: sns.boxplot(y="Profit",data=df,color="yellow")
```

```
[146]: <Axes: ylabel='Profit'>
```



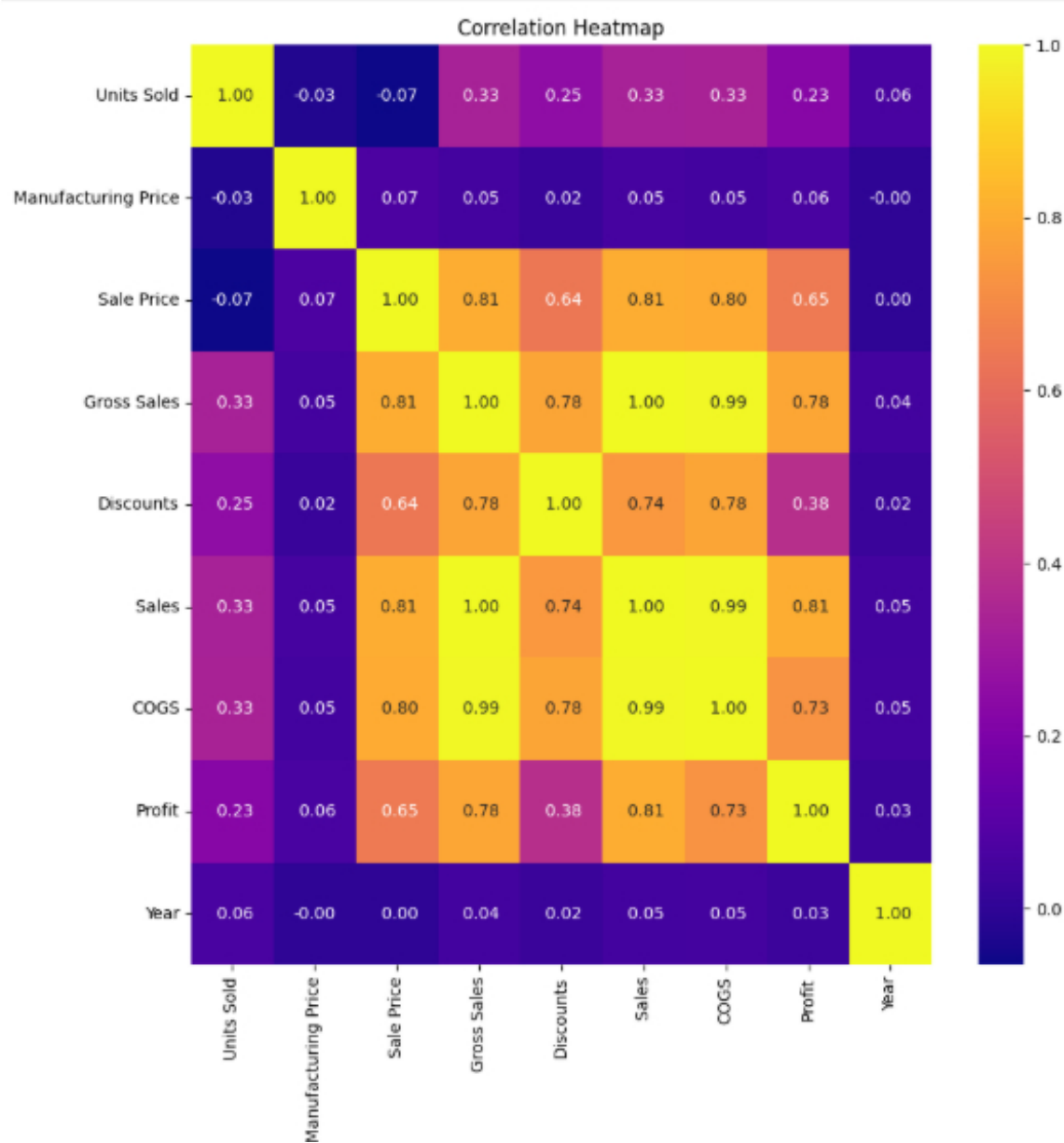
```
[147]: sns.boxplot(y="Units Sold",data=df,color="red")
```

```
[147]: <Axes: ylabel='Units Sold'>
```



What is the correlation between Units Sold, Sales, COGS, and Profit?

```
[141]: import matplotlib.pyplot as plt
import seaborn as sns
plt.figure(figsize=(18,18))
correlation = df.corr(numeric_only =True)
sns.heatmap (
    correlation,
    annot=True,
    cmap='plasma',
    fmt=".2f"
)
plt.title("Correlation Heatmap")
plt.show()
```



Create a regression model to predict sales

```
[150]: import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score

# Load dataset
df = pd.read_csv("Dataset.csv")

# Input and target
X = df[['Units Sold']]
y = df['Sales']

# Split data
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# Create and train model
model = LinearRegression()
model.fit(X_train, y_train)

# Predict sales
y_pred = model.predict(X_test)

# Check model performance
print("R2 Score:", r2_score(y_test, y_pred))
print("Mean Squared Error:", mean_squared_error(y_test, y_pred))

# Predict sales for a new quantity
new_quantity = [[10]]
prediction = model.predict(new_quantity)

print("Predicted Sales:", prediction[0])
```

R2 Score: 0.86535564527077464

Mean Squared Error: 51763524938.18301

Predicted Sales: 18040.775477365438

c:\users\krishnamohan b\appdata\local\programs\python\python39\lib\site-packages\sklearn\utils\validation.py:2739: UserWarning: X does not have valid feature names, but LinearRegression was fitted with feature names
warnings.warn(

Write a short summary of findings & recommendations.

Summary of Findings

The analysis reveals some significant differences in terms of sales and profits by product, year, country, and customer segment. Certain products contribute much more towards total sales and profit than others. Units sold are shown to have an important relationship with sales, although there may be impacts from discounts affecting profit margins. Analysis at a year- and month-wise level allows us to see periods of increased sales performance.

Recommendations

High performing products/customer segments should focus efforts upon to increase sales & profitability as required. It would seem prudent to monitor levels of discounting in order to prevent unwarranted reductions in profit margins. It will be necessary to analyze low performing products and regions in search of avenues of improvements. Sales Trends could assist planning for inventories and marketing activities and sales strategy's for high demand times.